Settings Page Info

Page Overview

Purpose

The Settings page is the central configuration surface for `resmon`. It groups all user-configurable behavior into eight sub-tabs — Email, Cloud Account, Cloud Storage, AI, Storage, Notifications, Advanced, and About App — each handling a self-contained slice of backend configuration, secret management, or diagnostics.

Primary User Flows

1. Open **Settings** from the sidebar. The page renders a tab strip and a panel area; the route defaults to the Email panel.
2. Click a tab to navigate to its panel (`/settings/email` , `/settings/account` , `/settings/cloud` , `/settings/ai` , `/settings/storage` , `/settings/notifications` , `/settings/advanced` , `/settings/about`).
3. Within a panel, edit fields and press **Save** (or the panel-specific action button) to persist the change through the corresponding `PUT /api/settings/*` endpoint (or a credential / service endpoint, depending on the tab).
4. Test actions (send test email, test API key, list models, link Google Drive, install service, refresh scheduler jobs) call their backend endpoints directly and surface an inline status line.

Inputs and Outputs

- **Inputs** — per-panel form fields, toggles, and action buttons; secrets typed into password inputs (SMTP password, AI provider API key).
- **Outputs** — rows written to the `settings` table via `set_setting` , OS-keychain entries written through `credential_manager.store_credential` , OAuth tokens acquired by `cloud_storage.authorize_google_drive` , and scheduler / service diagnostic data surfaced read-only.

Known Constraints

- Secrets (SMTP password, AI provider API keys) are never returned to the frontend; the UI only sees presence booleans from `GET /api/credentials` .
- The **Cloud Account** tab is an "under construction" placeholder in this build — the backend routes (IMPL-27..40) exist, but no hosted identity provider is wired, so sign-in is not available.
- The **AI** tab's Custom provider branch disables Save unless the base URL is HTTPS or the host is loopback (`localhost` , `127.0.0.1` , `::1`), per IMPL-AI12.
- The PDF / TXT retention policies on the **Storage** tab are reserved for a future per-paper artifact download feature and have no effect on current Deep Dive / Deep Sweep output.

Frontend

Route and Main Component

- **Route:** `/settings` (nested routes under `/settings/<tab>`).
- **Main component:** `SettingsPage` (resmon_scripts/frontend/src/pages/SettingsPage.tsx).

- The page renders an `<h1>Settings</h1>` header, a `.settings-nav` tab strip of `NavLink` elements, and a nested `<Routes>` block. The index route redirects to `/settings/email` via `<Navigate to="email" replace />`.

Child Components and Hooks

- Panel components (all under `resmon_scripts/frontend/src/components/Settings/`): `EmailSettings`, `CloudAccountSettings`, `CloudSettings`, `AISettings`, `StorageSettings`, `NotificationSettings`, `AdvancedSettings`, `AboutAppSettings`.
- Shared help component: `PageHelp` (each panel wires its own `storageKey`, e.g. `settings-email`, `settings-ai`, `settings-advanced`, `settings-about-app`).
- Hooks: `useState`, `useEffect`, `useCallback` for panel-local state; `NavLink` / `Navigate` / `Routes` from `react-router-dom` for tab routing; `apiClient` wrapper for REST calls.

UI State Model

`SettingsPage` itself holds no state. Each panel owns its own local state; common patterns:

- `loading: boolean` / `saving: boolean` — request guards.
- `status: string` — transient inline status line, cleared via `setTimeout`.
- A `settings` object mirroring the tab's settings subset (Email, AI, Storage, Notifications, Cloud Storage).
- Secret inputs (SMTP password, AI key) held in separate state with a companion `*Masked` flag; presence of the stored credential fetched as a boolean.

Key Interactions and Events

- **Tab switch** — `NavLink` updates the URL; the matching panel mounts and triggers its own initial `GET`.
- **Save** — each panel's primary button calls `apiClient.put('/api/settings/<slice>', { settings })` and reloads the presence / status it needs.
- **Test** — panel-specific side actions: `POST /api/settings/email/test`, `POST /api/credentials/validate`, `POST /api/ai/models`, `POST /api/cloud/backup`, `POST /api/service/install`.
- **Credential write / delete** — `PUT /api/credentials/{name}` and `DELETE /api/credentials/{name}` from Email and AI panels.

Error and Empty States

- Each panel shows a muted `Loading…` placeholder until its initial fetch resolves.
- Inline `status` strings surface both successes and errors; error strings are prefixed `Error:` / `Test failed` and auto-clear after a few seconds.
- The Cloud Account panel shows a fixed "Under construction" block instead of any functional control.
- Advanced shows `jobs=[]` with an error line if `GET /api/scheduler/jobs` fails, rather than hiding the diagnostics card.

Panel: Email

- Component: `EmailSettings` (resmon_scripts/frontend/src/components/Settings/EmailSettings.tsx).
- Fields: `smtp_server`, `smtp_port`, `smtp_username`, `smtp_from`, `smtp_to` — persisted through `PUT /api/settings/email`.
- SMTP password is stored in the OS keychain under the credential name `smtp_password` via `PUT /api/credentials/smtp_password`; the UI fetches only a presence boolean from `GET /api/credentials`. On store, whitespace is stripped from the input so Gmail App Passwords (displayed as four space-separated groups) become the raw 16-character secret.

- Actions: **Save**, **Store password**, **Remove password** (`DELETE /api/credentials/smtp_password`), **Send test email** (`POST /api/settings/email/test`).

Panel: Cloud Account

- Component: `CloudAccountSettings` (resmon_scripts/frontend/src/components/Settings/CloudAccountSettings.tsx).
- Renders a `PageHelp` block describing why cloud sign-in matters (cloud routines, cloud-scoped credentials, cloud-executed reports) and a privacy note that local executions never depend on cloud sign-in.
- Body is a static "Under construction" placeholder; no sign-in control is rendered because no hosted identity provider is wired in this build. The underlying backend routes (`/api/cloud-auth/*`) exist and are covered by the Backend section below.

Panel: Cloud Storage

- Component: `CloudSettings` (resmon_scripts/frontend/src/components/Settings/CloudSettings.tsx).
- State: `isLinked` , `apiOk` , `apiReason` , `autoBackup` — loaded from `GET /api/cloud/status` and `GET /api/settings/cloud` .
- Actions: **Link Google Drive** (`POST /api/cloud/link` — triggers the OAuth installed-app flow in the backend), **Unlink** (`POST /api/cloud/unlink`), **Toggle auto-backup** (writes `cloud_auto_backup` via `PUT /api/settings/cloud`), **Back up now** (`POST /api/cloud/backup` — returns `{ uploaded, total_files, folder_name, web_view_link }`).
- Surfaces Drive-API error reasons via `API_REASON_HINTS` (`accessNotConfigured` , `insufficientPermissions` , `no_token`) so the user can resolve Google Cloud Console / scope-consent issues.

Panel: AI

- Component: `AISettings` (resmon_scripts/frontend/src/components/Settings/AISettings.tsx).
- Provider whitelist (IMPL-AI9 / AI10): `anthropic` , `openai` , `google` , `xai` , `meta` , `deepseek` , `alibaba` , `local` , `custom` . Each provider has a suggested model placeholder (e.g. `gpt-4o-mini` , `claude-3-5-haiku-latest` , `gemini-2.5-flash` , `grok-4` , `meta-llama/Llama-3.3-70B-Instruct-Turbo` , `deepseek-chat` , `qwen-plus` , `llama3`).
- Settings keys persisted through `PUT /api/settings/ai` : `ai_provider` , `ai_model` , `ai_local_model` , `ai_summary_length` , `ai_tone` , `ai_temperature` , `ai_extraction_goals` , `ai_custom_base_url` , `ai_custom_header_prefix` . IMPL-AI11 separated `ai_local_model` from `ai_tone` and introduced a one-shot migration heuristic (`looksLikeModelId`) that moves a misplaced model id out of `ai_tone` .
- Custom-provider UX guard (IMPL-AI12): `validateCustomBaseUrl` disables Save if the base URL is not HTTPS unless the host is `localhost` , `127.0.0.1` , or `::1` . The backend `llm_factory` enforces the same rule as a hard check.
- Credential write: the provider's API key is stored via `PUT /api/credentials/{name}` where `name` is derived from the provider (e.g. `openai_api_key` , `anthropic_api_key` , `custom_llm_api_key`). The backend responds with presence only.
- Actions: **Save** (`PUT /api/settings/ai`), **Test key** (`POST /api/credentials/validate`), **Load models** (`POST /api/ai/models` — returns the per-provider model list using either the freshly typed key or the stored credential).

Panel: Storage

- Component: `StorageSettings` (resmon_scripts/frontend/src/components/Settings/StorageSettings.tsx).
- Fields: `pdf_policy` , `txt_policy` , `archive_after_days` , `export_directory` — persisted through `PUT /api/settings/storage` . Policy values are constrained to `save` / `archive` / `discard` .
- `export_directory` is the active setting used by configuration / execution exports. The PDF and TXT policy controls are marked in the UI as reserved for a future per-paper artifact download feature.
- Help copy calls out that retention policy prunes reports older than the window on daemon startup, and that the on-disk cloud-execution cache (`CLOUD_CACHE_MAX_BYTES_DEFAULT`) is capped independently.

Panel: Notifications

- Component: `NotificationSettings` (resmon_scripts/frontend/src/components/Settings/NotificationSettings.tsx).
- Fields: `notify_manual: boolean` and `notify_automatic_mode: 'all' | 'selected' | 'none'` — loaded from `GET /api/settings/notifications` and saved with `PUT /api/settings/notifications`.
- Also surfaces the browser-level desktop-notification permission state and a **Request permission** button that calls `Notification.requestPermission()` in the renderer.
- Help text clarifies that email notifications and cloud uploads are independent of this tab.

Panel: Advanced

- Component: `AdvancedSettings` (resmon_scripts/frontend/src/components/Settings/AdvancedSettings.tsx).
- **Background daemon** — `GET /api/service/status` returns `{ installed, unit_path, platform }`; `POST /api/service/install` and `POST /api/service/uninstall` toggle the OS service unit (IMPL-26). Health polling of `/api/health` at a 5-second interval surfaces `{ status, pid, started_at, version }`.
- **Concurrent executions (IMPL-R12)** — `GET /api/settings/execution` returns `{ max_concurrent_executions, routine_fire_queue_limit }`; the panel persists edits through `PUT /api/settings/execution`. Limits flow into `admission` (IMPL-R1 / R2) and into the scheduler's routine-fire queue (IMPL-R3 / R6).
- **Scheduler diagnostics** — `GET /api/scheduler/jobs` returns `{ id, name, next_run_time, trigger }` entries from the APScheduler job store. A Refresh action re-fetches the list.

Panel: About App

- Component: `AboutAppSettings` (resmon_scripts/frontend/src/components/Settings/AboutAppSettings.tsx).
- Displays app name / version (seeded to `1.0.0` and refreshed from `GET /api/health.version`, which is sourced from `implementation_scripts.config.APP_VERSION`), license and privacy summaries, and a set of author social links (GitHub, LinkedIn, X, website, email).
- Renders a copyright footer note. No mutations occur from this panel.

Backend

API Endpoints

Method	Path	Purpose	Panel
GET	`/api/health`	Daemon status `{ status, pid, started_at, version }`	About App, Advanced
GET	`/api/credentials`	Presence map for allowed credential names (no values)	Email, AI
PUT	`/api/credentials/{key_name}`	Store a secret via OS keyring	Email, AI
DELETE	`/api/credentials/{key_name}`	Remove a stored secret	Email, AI
POST	`/api/credentials/validate`	Probe an API key against its provider endpoint	AI
GET / PUT	`/api/settings/email`	SMTP configuration row set	Email
POST	`/api/settings/email/test`	Send a test email using the current config + stored password	Email

Method	Path	Purpose	Panel
GET / PUT	`/api/settings/ai`	AI provider / model / tone / temperature / extraction goals / custom-base-url settings	AI
POST	`/api/ai/models`	List models for a provider using ephemeral or stored key	AI
GET / PUT	`/api/settings/cloud`	Cloud-storage preferences (e.g. `cloud_auto_backup`)	Cloud Storage
POST	`/api/cloud/link`	Start Google Drive OAuth flow	Cloud Storage
POST	`/api/cloud/unlink`	Revoke and drop the Drive token	Cloud Storage
GET	`/api/cloud/status`	`{ is_linked, api_ok, api_reason }`	Cloud Storage
POST	`/api/cloud/backup`	Upload `resmon_reports/` to Drive; returns `{ uploaded, total_files, folder_name, web_view_link }`	Cloud Storage
GET / PUT	`/api/settings/storage`	PDF / TXT policies, archive retention, export directory	Storage
GET / PUT	`/api/settings/notifications`	`notify_manual` , `notify_automatic_mode`	Notifications
GET	`/api/service/status`	OS service-unit state	Advanced
POST	`/api/service/install`	Install platform-specific service unit	Advanced
POST	`/api/service/uninstall`	Remove platform-specific service unit	Advanced
GET / PUT	`/api/settings/execution`	`max_concurrent_executions` , `routine_fire_queue_limit`	Advanced
GET	`/api/scheduler/jobs`	APScheduler job listing	Advanced
POST / GET / DELETE / PUT	`/api/cloud-auth/session` , `/api/cloud-auth/status` , `/api/cloud-auth/refresh` , `/api/cloud-auth/sync`	Cloud account session plumbing (dormant UI in this build)	Cloud Account

Request / Response Patterns

- Settings reads (`GET /api/settings/<slice>`) return a flat `{ key: value }` mapping derived from `get_setting(conn, key)` calls in resmon_scripts/resmon.py.
- Settings writes use the envelope `PUT /api/settings/<slice> { "settings": { key: value, ... } }` ; the handler validates allowed keys for that slice and dispatches `set_setting(conn, key, value)` per entry.
- Credential routes never echo values. `GET /api/credentials` returns a `{ name: { present: bool } }` map constructed from the union of `AI_CREDENTIAL_NAMES ∪ SMTP_CREDENTIAL_NAMES ∪ catalog_credential_names()` .
- Validation (`POST /api/credentials/validate`) invokes `credential_manager.validate_api_key(provider, key, base_url?)` , which is intentionally non-raising — any non-200 response becomes `False` .

- Model listing (`POST /api/ai/models`) delegates to `ai_models.list_available_models` , which raises `ModelListError` on auth / transport failure.

Persistence Touchpoints

- `settings` table (SQLite) via `get_setting` / `set_setting` for every `/api/settings/*` endpoint. Schema version tracked in `SCHEMA_VERSION` .
- OS keychain via `credential_manager.store_credential` / `get_credential` / `delete_credential` (Python `keyring` , per ADQ-4). Service name is `resmon` (`APP_NAME`).
- Google Drive OAuth token persisted through `cloud_storage.authorize_google_drive` / `revoke_authorization` ; presence probed by `cloud_is_token_stored` and API reachability by `cloud_probe_api` .
- APScheduler job store (SQLAlchemy over SQLite, per ADQ-3) queried by `GET /api/scheduler/jobs` .
- OS-level service unit files (launchd / systemd / Task Scheduler) written by the `/api/service/install` path.

Execution Side Effects

- Saving `/api/settings/execution` updates the in-process `admission` controller's `max_concurrent_executions` and the scheduler's `routine_fire_queue_limit` , affecting subsequent manual and routine launches.
- `POST /api/settings/email/test` sends a real SMTP message using the stored `smtp_password` .
- `POST /api/cloud/backup` walks `REPORTS_DIR` and uploads files to Google Drive, returning the created folder's share link.
- `POST /api/service/install` and `/uninstall` mutate system-level service units and therefore whether routines can run while the app is closed.
- `POST /api/credentials/validate` and `/api/ai/models` issue a small number of outbound HTTPS requests to the provider's models endpoint.

Endpoints: Email Panel

- `GET /api/settings/email` — returns `{ smtp_server, smtp_port, smtp_username, smtp_from, smtp_to }` .
- `PUT /api/settings/email` — writes the same keys.
- `POST /api/settings/email/test` — constructs and sends a test message via the configured SMTP server using `smtp_password` from the keychain.
- `PUT /api/credentials/smtp_password` / `DELETE /api/credentials/smtp_password` — manages the keychain entry.

Endpoints: Cloud Account Panel

- `POST /api/cloud-auth/session` , `GET /api/cloud-auth/status` , `DELETE /api/cloud-auth/session` , `POST /api/cloud-auth/refresh` , `PUT /api/cloud-auth/sync` — JWKS-verified session lifecycle endpoints per IMPL-29 / 30. Present in the backend but not currently exercised by this panel because no hosted identity provider is wired.

Endpoints: Cloud Storage Panel

- `GET /api/cloud/status` — `{ is_linked, api_ok, api_reason }` .
- `POST /api/cloud/link` — runs `authorize_google_drive` (installed-app OAuth with `drive.file` scope).
- `POST /api/cloud/unlink` — runs `revoke_authorization` and discards the stored token.
- `GET / PUT /api/settings/cloud` — reads and writes `cloud_auto_backup` (and any other cloud-storage-scoped settings).
- `POST /api/cloud/backup` — walks `REPORTS_DIR` with `cloud_storage.upload_directory` and returns `{ uploaded, total_files, folder_name, web_view_link }` .

Endpoints: AI Panel

- `GET /api/settings/ai` / `PUT /api/settings/ai` — reads and writes the AI settings keys (see panel section above).
- `POST /api/credentials/validate` — `credential_manager.validate_api_key(provider, key, base_url?)`.
- `POST /api/ai/models` — `ai_models.list_available_models` using either an ephemeral key (pushed with `push_ephemeral` for the current request) or the stored credential.
- `PUT /api/credentials/{name}` / `DELETE /api/credentials/{name}` where `name ∈ AI_CREDENTIAL_NAMES` (`openai_api_key`, `anthropic_api_key`, `google_api_key`, `xai_api_key`, `meta_api_key`, `deepseek_api_key`, `alibaba_api_key`, `custom_llm_api_key`).

Endpoints: Storage Panel

- `GET /api/settings/storage` — returns `{ pdf_policy, txt_policy, archive_after_days, export_directory }`.
- `PUT /api/settings/storage` — writes the same keys; backend coerces numeric `archive_after_days` and validates policy values against `{save, archive, discard}`.

Endpoints: Notifications Panel

- `GET /api/settings/notifications` — returns `{ notify_manual, notify_automatic_mode }`.
- `PUT /api/settings/notifications` — writes the same two keys; `notify_automatic_mode` is constrained to `'all' | 'selected' | 'none'`.

Endpoints: Advanced Panel

- `GET /api/service/status`, `POST /api/service/install`, `POST /api/service/uninstall` — OS service-unit management (IMPL-26).
- `GET /api/settings/execution` / `PUT /api/settings/execution` — `{ max_concurrent_executions, routine_fire_queue_limit }` used by `admission` (IMPL-R1) and the scheduler's routine-fire queue (IMPL-R3 / R6).
- `GET /api/scheduler/jobs` — APScheduler job listing for diagnostics.
- `GET /api/health` — polled every 5 s to display PID, uptime, and version.

Endpoints: About App Panel

- `GET /api/health` — supplies the `version` field displayed on the panel. The daemon populates this from `implementation_scripts.config.APP_VERSION`; `APP_NAME` is the same constant used as the keyring service name.

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